Continual Learning Benchmark: Comparative Study of Anti-Forgetting Methods on Split CIFAR-100

Hammani Abdeslem
Fellah Mahdi
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Ecole nationale Supérieure d'Informatique (ESI), Algiers

1 Introduction

Continual learning (CL) aims to train neural networks on a sequence of tasks without *catastrophic forgetting*—the tendency to lose previously acquired knowledge when learning new information [1]. This is a fundamental obstacle to building systems that learn incrementally. In this work, we compare six anti-forgetting methods—spanning regularization (EWC, SI), distillation (LwF), and replay (ER, GEM, DER++)—against a naïve fine-tuning baseline on Split CIFAR-100. We evaluate using average accuracy, forgetting, BWT, and FWT following [6].

2 Dataset & Experimental Setup

2.1 Dataset & Task Split

We use **CIFAR-100** (60 000 images, 100 classes, 32×32 RGB). The 100 classes are divided into $K = 10$ sequential tasks of 10 classes each, with a fixed random class permutation (seed 42) for reproducibility. Each task contains 5 000 training and 1 000 test images.

2.2 Model Architecture

All methods share a **CNN backbone** with 3 convolutional blocks (32, 64, 128 filters), each with two Conv–BN–ReLU layers, MaxPool, and Dropout. The classifier consists of FC(2048→256)–ReLU–Dropout–FC(256→100). Total: $\sim$700K parameters.

2.3 Methods Compared

- **Fine-Tune** (baseline): Sequential CE training, no forgetting protection.
- **EWC** [2]: Quadratic penalty weighted by diagonal Fisher information.
- **SI** [3]: Online path-integral importance tracking.
- **LwF** [4]: KL-divergence distillation from a frozen teacher.
- **ER** [5]: Reservoir sampling replay interleaved with current data.
- **GEM** [6]: A-GEM gradient projection using episodic memory.
- **DER++** [7]: Replay with stored logit distillation (dark knowledge).

2.4 Hyperparameters

Training: SGD (lr= 0.01, momentum= 0.9, weight decay= 10^{-4}), batch size 64, 10 epochs/task, task-aware evaluation (output head masking).

Table 1: Method-specific hyperparameters.

Param	EWC	SI	LwF	ER	GEM	DER++
λ / c	400	1.0	1.0	–	–	–
Buffer M	–	–	–	500	50/task	500
T / α, β	–	–	2.0	–	–	0.5, 0.5

3 Results

3.1 Accuracy Matrix

Figure 1 shows the full 10×10 accuracy matrix $A_{i,j}$ for each method, where $A_{i,j}$ is the accuracy on task j after training on task i. The diagonal captures accuracy immediately after training; the lower-left triangle reveals forgetting. Fine-Tune and GEM show near-zero off-diagonal entries (severe forgetting), whereas DER++ and ER maintain high values across all columns.

3.2 Metric Definitions

We report four standard metrics from the accuracy matrix A:

$$\bar{A} = \frac{1}{K}\sum_{j=1}^{K} A_{K,j} \quad (1)$$

$$\bar{F} = \frac{1}{K-1}\sum_{j=1}^{K-1}\left(\max_i A_{i,j} - A_{K,j}\right) \quad (2)$$

$$\mathrm{BWT} = \frac{1}{K-1}\sum_{j=1}^{K-1}(A_{K,j} - A_{j,j}) \quad (3)$$

$$\mathrm{FWT} = \frac{1}{K-1}\sum_{j=2}^{K}(A_{j-1,j} - b_j) \quad (4)$$

where $b_j = 10\%$ is the random baseline for 10-class tasks.

3.3 Summary Table

Table 2: Quantitative comparison on Split CIFAR-100 ($K = 10$). ↑/↓ indicate desirable direction.

Method	$\bar{A}$↑	$\bar{F}$↓	BWT	FWT
Fine-Tune	18.17	48.01	−48.01	−0.09
EWC	42.00	16.10	−16.10	+1.04
SI	10.00	13.17	−11.70	+0.89
LwF	32.80	41.36	−41.36	−0.91
ER	58.05	13.32	−11.61	−0.44
GEM	20.77	49.01	−49.01	+1.24
DER++	**61.83**	**15.70**	−15.70	+0.32

4 Analysis & Discussion

Replay vs. regularization. ER ($\bar{A} = 58.1\%$) and DER++ ($\bar{A} = 61.8\%$) strongly outperform EWC (42.0%)

and LwF (32.8%), confirming that 500 exemplars is more effective than parameter constraints. DER++ achieves the best result by combining replay with logit distillation ("dark knowledge").

BWT and forgetting. Fine-Tune has BWT$=-48.0$ vs. EWC's -16.1 vs. ER's -11.6. EWC reduces forgetting $3\times$ via Fisher-weighted penalties. SI achieves low $\bar{F}=13.2$ but collapses to $\bar{A}=10\%$, indicating $c=1.0$ over-constrains learning.

Surprising results. GEM ($\bar{A}=20.8\%$, BWT$=-49.0$) underperforms Fine-Tune—50 samples/task yields noisy reference gradients for A-GEM projection. LwF underperforms EWC as distillation degrades when CIFAR-100 subsets are dissimilar [4]. FWT≈ 0 for all methods, confirming minimal inter-task transfer.

Limitations. (1) Single seed; 3–5 runs recommended. (2) Task-aware evaluation; class-incremental is harder. (3) Shallow CNN; deeper models may alter rankings.

5 Conclusion

DER++ offers the best stability-plasticity trade-off ($\bar{A}=61.8\%$), combining replay with logit distillation. Replay-based methods consistently dominate regularization and distillation approaches on Split CIFAR-100 [7, 8]. Among non-replay methods, EWC provides the strongest forgetting reduction. A natural extension is to combine replay with regularization, or to evaluate under the harder class-incremental protocol.

References

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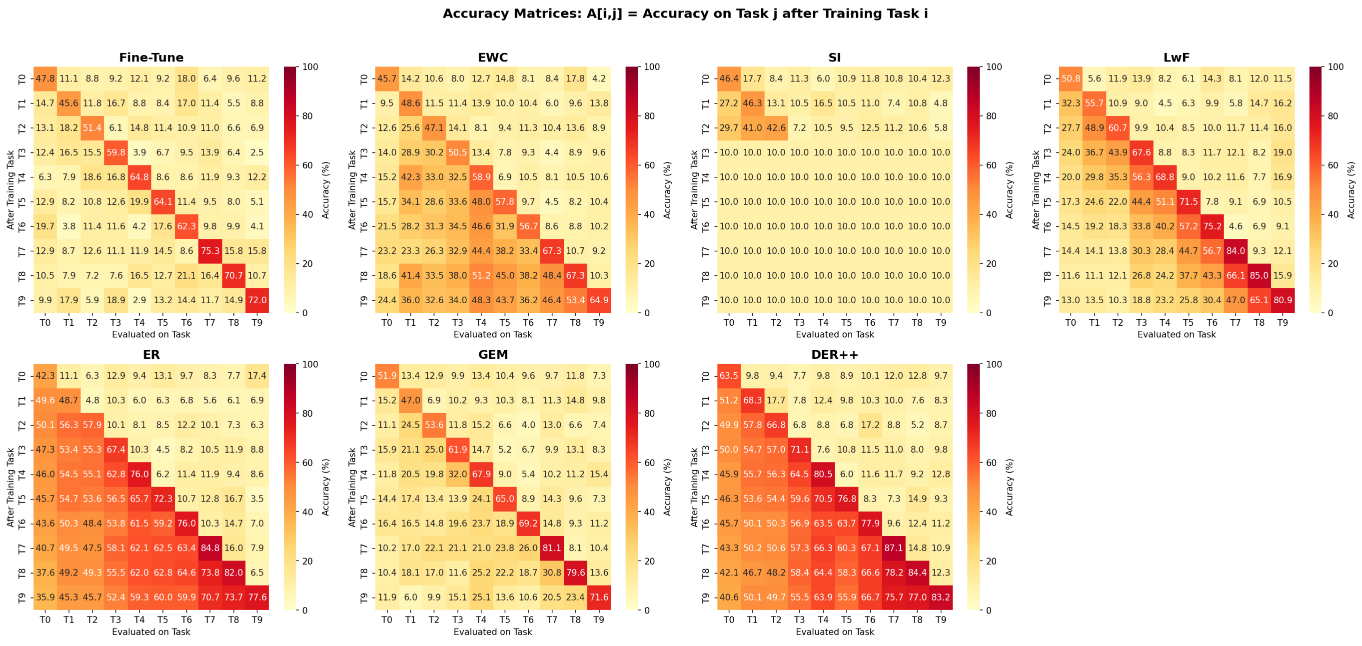

Figure 1: Accuracy matrices $A_{i,j}$ for all seven methods. Colour scale fixed to $[0, 100]$. DER++ and ER maintain the highest off-diagonal accuracies; Fine-Tune and GEM exhibit severe forgetting.